BIOCHEMISTRY CUMULATIVE EXAMINATION

October 6, 1990

You are studying the action of a hormone on cell metabolism. The hormone stimulates the production of glucose-1-phosphate from glycogen. You have a tritiated version of the hormone available.

(25) 1. You incubate cells with the tritiated hormone and then want to see whether the hormone binds to the cell membrane, to the nucleus, or to a soluble cytoplasmic protein. How would you separate these components?

(25) 2. You find that the hormone binds to cell membranes. You treat a preparation of cell membranes with the hormone to see if the production of glucose-1-phosphate is stimulated in the presence of exogenous glycogen. How will you conduct this assay?

(25) 3. In part 2, the result turns out negative; no glucose-1-phosphate is produced. If, however, a soluble cytoplasmic fraction is added to the membrane preparation in the presence of glycogen, then the addition of hormone will stimulate production of glucose-1-phosphate. Propose a mechanism of action for the hormone. How could you further test the mechanism?

(25) 4. You obtain saturation binding data that suggest only 20 copies of the receptor molcule per cell. Given its low abundance, how would you go about trying to purify the receptor? What assay would you use?